

Mechanistic insights into the effect of CYP2C9*2 and CYP2C9*3 variants on the 7-hydroxylation of warfarin

Aim: To evaluate the impact of CYP2C9*2 and CYP2C9*3 variants on binding and hydroxylation of warfarin. **Materials & methods:** Multiple linear regression model of warfarin pharmacokinetics was developed from the dataset of patients (n = 199). Pymol based *in silico* models were developed for the genetic variants. **Results:** CYP2C9*2 and CYP2C9*3 variants exhibited high warfarin/7-hydroxywarfarin (multiple linear regression model), dose-dependent disruption of hydrogen bonds with warfarin, dose-dependent increase in the distance between C7 of S-warfarin and Fe–O of CYP2C9, dose-dependent decrease in the glide scores (*in silico*). **Conclusion:** CYP2C9*2 and CYP2C9*3 variants result in disruption of hydrogen bonding interactions with warfarin and longer distance between C7 and Fe–O thus impairing warfarin 7-hydroxylation due to lower binding affinity of warfarin.

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Keywords: 7-hydroxy warfarin • CYP2C9*2 • CYP2C9*3 • multiple linear regression • warfarin

Background

CYP2C9 (MIM:*601130) is one of the major drug-metabolizing enzyme of the CYP450 superfamily of monooxygenases, which is involved in the metabolism of more than 100 therapeutic drugs [1]. CYP2C9 gene is highly polymorphic with several SNPs reported in its regulatory and coding regions. Some of these SNPs are associated with reduced enzyme activity compared with wild-type both *in vitro* and *in vivo* [2–5].

Warfarin is an oral anticoagulant frequently prescribed to prevent thrombotic events in patients with deep vein thrombosis, atrial fibrillation and valve replacement therapy. Warfarin is a racemic mixture of R- and S-enantiomers. S-warfarin is more potent than R-warfarin in producing an anticoagulant response [6]. CYP2C9 catalyzes the 7-hydroxylation of S-warfarin to form 7-hydroxy warfarin [6]. Two polymorphisms in CYP2C9, that is, CYP2C9*2 (rs1799853) and CYP2C9*3 (rs1057910)

were reported to decrease CYP2C9 activity [7]. Adverse drug reactions were observed in the subjects harboring these two polymorphisms upon treatment with the standard empirical doses of warfarin [8]. Typically, they exhibit a phenomenon termed as ‘warfarin sensitivity,’ requiring lower doses of warfarin for achieving therapeutic international normalized ratio (INR) due to poor 7-hydroxylation [9].

The CYP2C9*2 and CYP2C9*3 variants are being used widely as predictors of warfarin dose in different pharmacogenomic algorithms [10,11]. Our recent studies also highlighted potential utility of these genetic variants in warfarin dose optimization [12,13]. These polymorphisms result in the following amino acid substitutions, that is, R144C and I359L, respectively. Although, these amino acids are located away from the active catalytic site of CYP2C9, they are influencing S-warfarin metabolism significantly. The existing simulation models

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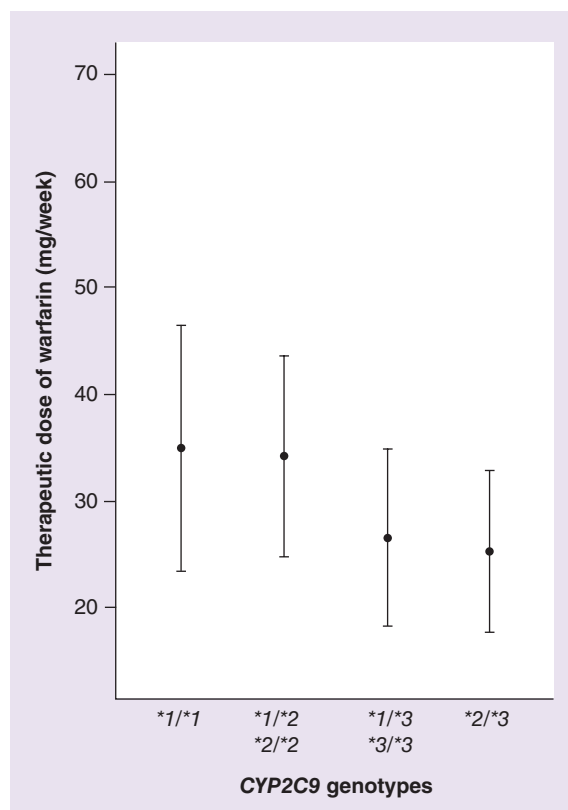


Figure 1. Impact of *CYP2C9* genetic polymorphisms on the therapeutic dose of warfarin. Analysis of variance was performed to study the impact of the *CYP2C9* genotypes on the therapeutic dose of warfarin. The black dot represents mean and the bars represent standard deviation. This analysis showed significant lowering of therapeutic doses of warfarin in *CYP2C91/*2, *1/*3 and *CYP2C9**2/*3 genotypes (F: 4.08, p = 0.009).**

proposed following molecular effects of *CYP2C9**2 and *CYP2C9**3 variants: an increased fluctuation of residues near the substrate recognition site/active site [14], reduction in the substrate access channel size hindering the entry of substrate to the active site [15] and increased distance between hydroxylation site of S-warfarin and reactive oxyferryl heme than wild-type structure [16].

In the current study, we have generated PyMOL based structural ensembles of each variant using crystal structure of CYP2C9 with flurbiprofen bound form (PDB ID: 1R9O) as a template. This has better resolution (2 Å) compared with other two crystal structures of CYP2C9, that is, 1OG2 (2.60 Å) and 1OG5 (2.55 Å). Glide (grid-based ligand docking with energetics) score was used for docking and binding affinity prediction in wild and variant forms of CYP2C9 with S-warfarin to provide more insights into the CYP2C9 mediated molecular dynamics and warfarin pharmacokinetics.

Materials & methods

Patients & data collection

A total of 199 patients (110 men and 89 women) who are receiving therapeutic dose of warfarin as a preventive measure for thrombotic events for clinical indications, that is, valve replacement (n = 195), deep vein thrombosis (n = 2) and atrial fibrillation (n = 2) were enrolled as the study subjects. INR values were used as the indices of therapeutic efficacy. All the patients who had INR values in the range of 2.0–3.5 with prolonged time in therapeutic range (>50–100%) were included in the derivation cohort (n = 123). Cases with hypothyroidism were excluded from the analysis (n = 9). The study protocol was approved by the ethical committee of Nizam's Institute of Medical Sciences, Hyderabad (EC/NIMS/1170/2010). Informed consent was obtained from all the patients.

Genetic analysis

Whole blood samples were collected in ethylenediaminetetraacetic acid and genomic DNA was isolated using phenol-chloroform extraction method. *CYP2C9**2 and *CYP2C9**3 polymorphisms were analyzed by PCR-RFLP method as described earlier [12].

Analysis of warfarin & 7-hydroxy warfarin in clinical samples

Plasma sample of the patients undergoing warfarin therapy were analyzed for warfarin and 7-hydroxy-warfarin levels using reverse phase HPLC as described earlier [17].

Molecular modeling

UNIPROT database was referred for information regarding the impact of *CYP2C9**2 and *CYP2C9**3 polymorphisms on the protein structure. The crystal structure of CYP2C9 with flurbiprofen bound form (PDB ID: 1R9O) was used as a template for constructing substrate-free forms of *CYP2C9**1 (wild) and mutants (R144C, I359L and I359T). The figures were generated by PyMOL [18]. Glide module of Schrodinger was used for removing flurbiprofen and docking of S-warfarin with wild-type CYP2C9, *CYP2C9**2, *CYP2C9**3, *CYP2C9**2/*3. The protein was prepared using ProteinPrep wizard. All the lone water molecules and other heteroatoms, except flurbiprofen were removed. The structure was optimized with neutral pH and restrained minimization was done to stabilize the protein. The grid was centered on the ligand flurbiprofen. The docking was performed using XP precision.

Using 3D structure of (S)-warfarin bound CYP2C9 as a reference, docking for a putative catalytic conformation of (S)-warfarin in the active-site

of *CYP2C9*1* was done that could yield a major metabolite 7-hydroxywarfarin, using Glide module of Schrodinger.

Statistical analysis

Analysis of variance was performed by computing the therapeutic dose of warfarin across four genotype combinations, that is, *CYP2C9*1* (wild), *CYP2C9*2*, *CYP2C9*3* and *CYP2C9*2/*3*. Multiple linear regression analysis was performed using age, gender, BMI, *CYP2C9*2* and *CYP2C9*3* variants as predictor variables and warfarin/7-hydroxywarfarin ratio as outcome variable. A computational software www.xuru.org was used for this analysis.

Results

In the derivation cohort, 80 patients had *CYP2C9*1/*1* genotype. Nineteen patients had *CYP2C9*1/*2* genotype and 17 had *CYP2C9*1/*3* genotype. One patient was homozygous mutant for *CYP2C9*2*, that is, **2/*2* and one patient was homozygous mutant for *CYP2C9*3*, that is, **3/*3*. Five patients were compound heterozygous for both the variants, that is, **2/*3*.

As shown in Figure 1, The mean $\pm$ standard deviation of therapeutic doses of warfarin according to *CYP2C9* genotypes were as follows: *CYP2C9*1* = 34.9 ± 11.5 mg/week; *CYP2C9*2* = 34.1 ± 9.4 mg/week; *CYP2C9*3* = 26.4 ± 8.3 mg/week and *CYP2C9*2/*3* = 25.2 ± 7.6 mg/week.

The plasma samples of the patients on stable anti-coagulation were analyzed for warfarin and 7-hydroxy warfarin levels. Univariate analysis revealed age and gender as the most relevant predictors of warfarin/7-OH warfarin ratio followed by genetic factors (Table 1). The mean $\pm$ standard deviation of warfarin/7-hydroxy warfarin ratios according to *CYP2C9* genotypes were as follows: *CYP2C9*1/*1* = 0.52 ± 0.45 ; *CYP2C9*2* (**1/*2* & **2/*2*) = 0.62 ± 0.50 ; *CYP2C9*3* (**1/*3* & **3/*3*) = 0.65 ± 0.51 and *CYP2C9*2/*3* = 1.60 ± 1.11 ($P_{ANOVA} = 0.0006$).

The following multiple linear regression equation was deduced from this data to explain the contribu-

tion of demographic and genetic variables on the pharmacokinetics of warfarin, which contributed to 36% variability:

$$\text{Warfarin/7-hydroxy warfarin} = \text{antilog} ([0.007999 \times \text{age in year}] + [0.3126 \times \text{gender}] + [0.01222 \times \text{BMI in kg/m}^2] + [0.01143 \times \text{therapeutic dose in mg/week}] + [0.2072 \times \text{CYP2C9*2}] + [0.2802 \times \text{CYP2C9*3}] - [1.610891769]).$$

The experimental data and the predicted data showed good correlation ($r = 0.36$). This multiple linear regression (MLR) model showed good accuracy in predicting warfarin/7-OH ratio based on the demographic and genetic parameters (overall accuracy: 0.82, 95% CI: 0.73–0.893). This model has good specificity (odds ratio: 0.86, 95% CI: 0.76–0.92) and moderate sensitivity (odds ratio: 0.79, 95% CI: 0.71–0.86). Rate of overestimation and rate of underestimation of this model were 14.8 and 20.3%, respectively.

The PyMOL model of *CYP2C9*1* (Figure 2A) was docked with S-warfarin using Glide Schrodinger module and the protein–ligand interactions were elucidated. Warfarin was found to form two hydrogen bonds with protein backbone, that is, I205 and S209; and one hydrogen bond with protein side chain, that is, T301 in the wild-type of *CYP2C9*. π – π stacking interaction was also observed between the benzene ring of warfarin and F100 (Figure 3A). The distance between the FeO and the C7 was 5.80 Å (Figure 4A).

In *CYP2C9*2* variant (Figure 2B), warfarin formed only one hydrogen bond with the protein side chain, that is, R132 and exhibited stacking interaction between its benzene ring and F134 (Figure 3B). The distance between the FeO and the C7 in *CYP2C9*2* was 7.02 Å (Figure 4B).

*CYP2C9*3* (Figure 2C) resulted in a variant that exhibited only one hydrogen bond with the protein side chain (Figure 3C). The distance between the FeO and C7 was found to be 7.46 Å (Figure 4C).

In *CYP2C9*2/*3* double variant (Figure 2D), the ligand exhibited stacking interaction with F100 using its benzene ring (Figure 3D). As observed, the distance between the FeO and the C7 was about 10.07 Å

Table 1. Univariate linear regression analysis showing association of demographic and genetic variables with warfarin/7-OH warfarin ratio.

Parameter	Warfarin/7-OH warfarin prediction	Correlation coefficient
Age	$(0.0142 \times \text{age in years}) + 0.256$	0.197
Gender	$(0.4317 \times \text{male gender}) + 0.556$	0.198
BMI	$(0.0128 \times \text{BMI in kg/m}^2) + 0.522$	0.047
Therapeutic dose	$(0.0174 \times \text{warfarin dose in mg/week}) + 0.205$	0.178
<i>CYP2C9*2</i>	$(0.2729 \times \text{number of *2 alleles}) + 0.738$	0.108
<i>CYP2C9*3</i>	$(0.3363 \times \text{number of *3 alleles}) + 0.668$	0.168

(Figure 4D), which is greater than all other genotypes. The glide scores (Kcal/mol) of wild, *CYP2C9*2*, *CYP2C9*3* and *CYP2C9*2/*3* were -7.698, -7.380, -6.821 and -6.986, respectively.

While performing structural analysis between the docked models generated with warfarin bound to *CYP2C9* and already available crystal structure (1OG5), it was found that warfarin binds nearly at the same position as in the crystal structure and it has an RMSD of 0.608 Å.

Discussion

In the current study, both *in vivo* and *in silico* approaches were used to understand the impact of *CYP2C9*2* and *CYP2C9*3* variants on the pharmacokinetics of warfarin. *In vivo* data showed lower therapeutic dose of warfarin in these variants compared with wild genotypes which is consistent with the existing literature [19,20]. The multiple linear regression model showed that these variants are associated with higher warfarin/7-hydroxy warfarin ratio suggesting poor metabolism of warfarin. This is consistent with the S-warfarin clearance data showing reduction of clearance of S-warfarin

in the presence of *CYP2C9*2* and *CYP2C9*3* variants [21]. The developed model had good accuracy, greater specificity, moderate sensitivity and lower rates of overestimation and underestimation. Further, this model explained previously reported associations of demographic variables in metabolic perspective [12]. With increase in age, decrease in the metabolism of warfarin was observed. With increase in BMI, there was decrease in warfarin metabolism.

In silico analysis was conducted to study the protein–ligand interactions, to measure the distance between FeO of the protein and the C7 of the ligand and to study the affinity of warfarin toward *CYP2C9*. Greater the number of the protein–ligand interactions and lesser the distance between FeO and C7, greater will be the metabolic activity of the *CYP2C9* enzyme.

The two hydrogen bonds of warfarin with protein backbone were disrupted in the presence of *CYP2C9*2* and *CYP2C9*3* variants while all the three hydrogen bonds were disrupted in double variant. The amino acid substitutions at both the positions (R144C and I359L) simultaneously caused a great conformation change to the protein resulting in the loss of all the

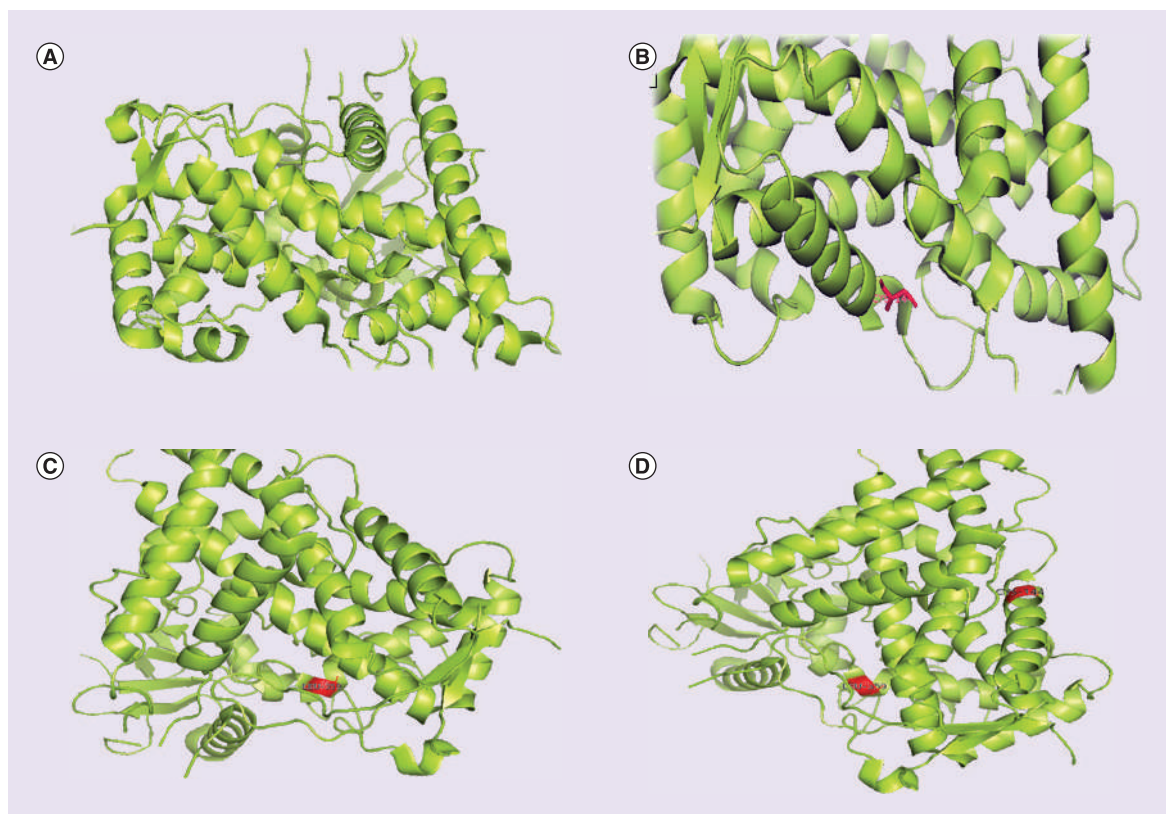


Figure 2. 3D structures of *CYP2C9* wild and variant enzymes. Using the crystal structure of *CYP2C9* with flurbiprofen bound form (PDB ID: 1R9O) as a template, substrate-free forms of (A) *CYP2C9*1*, (B) *CYP2C9*2*, (C) *CYP2C9*3* and (D) *CYP2C9*2/*3* were generated using Pymol. Red color represents place of change in the protein structure.

For color image please see www.futuremedicine.com/doi/pdf/10.2217/pgs.14.185

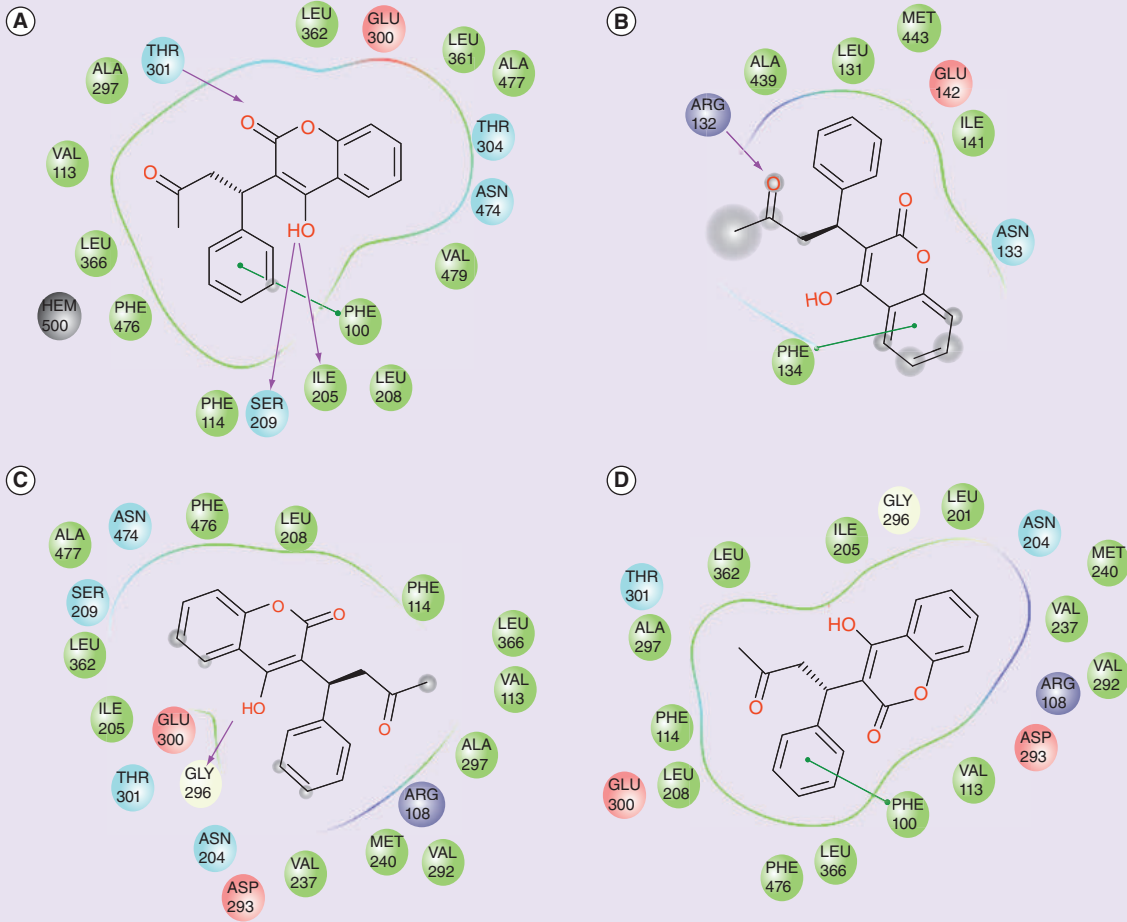


Figure 3. Protein–ligand interactions with reference to CYP2C9 variants. (A) Warfarin showing two hydrogen bonds with protein backbone, one hydrogen bond with the protein side chain and one π - π interaction with F100 of the wild-type of protein (CYP2C9*1). (B) Warfarin exhibits only one hydrogen bond with the protein side chain and one π - π interaction in CYP2C9*2. (C) Warfarin exhibits only one hydrogen bond with the protein side chain and one π - π interaction in CYP2C9*3. (D) Warfarin exhibits no hydrogen bonding with the protein and π - π interaction is retained in CYP2C9*2/*3.

major protein–ligand interactions. Another contributory factor for the decreased binding affinity of warfarin in *CYP2C9**2 variant was reported to be disruption of hydrogen bond network of R144 hydrogen with carbonyl groups of M136, R139, S140 on the loop between C-helix and D-helix, and Q261 on the loop between G-helix and H-helix [16].

Quantum mechanics/molecular mechanics modeling suggested C–O bond formation between the C6 and C7 of S-warfarin and the Cpd I ferryl oxygen of CYP2C9 with relative preference for C7 [22]. When compared with *CYP2C9*1*, there was a gradual increase in the distance between FeO and C7 in *CYP2C9*2*, *CYP2C9*3* and *CYP2C9*2/*3* variants. The glide scores suggest decreased affinity of warfarin in the *CYP2C9*3* and *CYP2C9*2/*3* variants compared with the wild-type protein.

The results of the current study corroborated with the meta-analysis of the enzyme activities of CYP2C9 variants that showed mean decreases in S-warfarin clearance of 15 and 45% compared with wild-type activity, respectively, for every *2 and *3 allele and values of unbound oral clearance of (S)-warfarin were predicted to be 19.9 (*1/*1), 17.1 (*1/*2), 10.8 (*1/*3), 14.8 (*2/*2), 7.6 (*2/*3) and 1.9 (*3/*3) l h⁻¹ [23]. Furthermore, CYP2C9*2 was reported to exhibit a rate of NADPH consumption similar to CYP2C9*1, while CYP2C9*3 exhibited lowest consumption of NADPH [24].

The current study has certain limitations. The HPLC analysis performed cannot distinguish enantiomer-specific metabolism of warfarin. The variables in the proposed multiple linear regression model were able to explain only 36% variability in warfarin pharmacokinetics. This could be due to unreported variants of

CYP2C9 and other CYP450 enzymes involved in the metabolism of R-warfarin. Although bioinformatic tools might be of help in understanding protein structure and interactions, they are not accurate enough to reach definitive conclusions especially for large molecules such as *CYP2C9*.

Conclusion

To conclude, the MLR model developed in this study has good accuracy, greater specificity, moderate sensitivity, lower rate of underestimation and overestimation in predicting the pharmacokinetics of warfarin. This can be further strengthened by including other genetic polymorphisms of *CYP2C9*. This model will help in predicting the rate of conversion of warfarin to 7-hydroxywarfarin and hence may help in ensuring safety and efficacy associated with warfarin therapy. *In silico* models provide

new insights into the interactions between the *CYP2C9* and warfarin highlighting the necessity of hydrogen bonding interactions between the protein and the ligand for increased affinity and efficient 7-hydroxylation. These interatomic interactions reduce the distance between the FeO and the C7. In *CYP2C9*3* and *CYP2C9*2/*3* variants, these interactions were lost resulting in reduced affinity to the warfarin and poor hydroxylation.

Future perspective

The MLR model developed in this study can be further strengthened by including other genetic polymorphisms of *CYP2C9* to understand the pharmacokinetics of warfarin. This model will help in predicting the rate of conversion of warfarin to 7-hydroxywarfarin and hence may help in ensuring safety and efficacy associated with warfarin therapy.

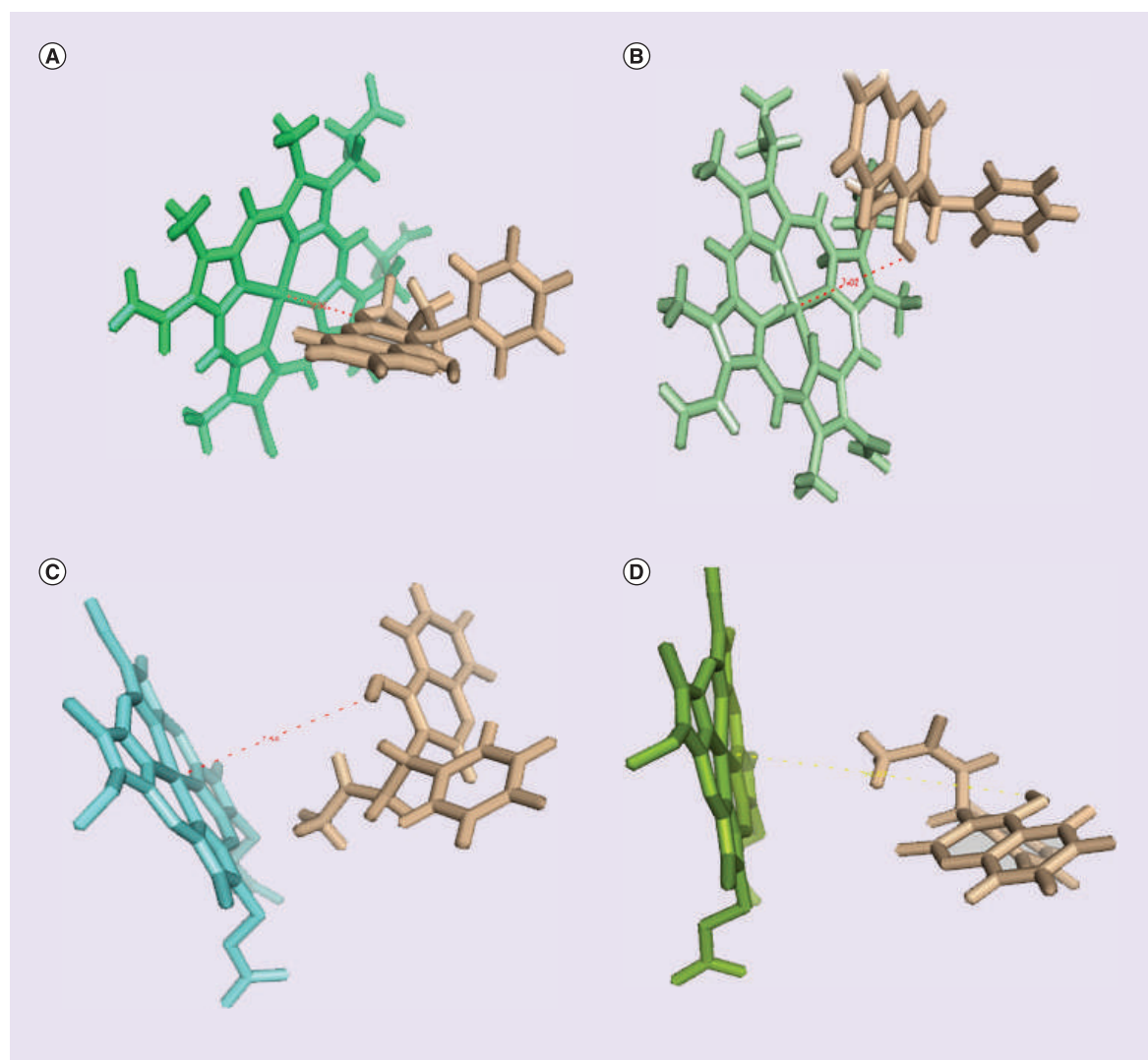


Figure 4. The distance plots between oxyferryl group of *CYP2C9* and C7 of warfarin. The distances between C7 of S-warfarin and Fe-O in (A) *CYP2C9*1*, (B) *CYP2C9*2*, (C) *CYP2C9*3* and (D) *CYP2C9*2/*3* were 5.81 Å, 7.02 Å, 7.43 Å and 10.07 Å, respectively.

Financial & competing interests disclosure

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No writing assistance was utilized in the production of this manuscript.

Ethical conduct of research

The authors state that they have obtained appropriate institutional review board approval or have followed the principles outlined in the Declaration of Helsinki for all human or animal experimental investigations. In addition, for investigations involving human subjects, informed consent has been obtained from the participants involved.

Executive summary**Background**

- CYP2C9 catalyzes 7-hydroxylation of warfarin.
- CYP2C9*2 and CYP2C9*3 polymorphisms impair this 7-hydroxylation and hence lower doses of warfarin are sufficient for effective anticoagulation.
- This study provides metabolic and mechanistic perspective on the poor metabolizer phenotype associated with CYP2C9*2 and CYP2C9*3 polymorphisms.

Materials & methods

- Patients receiving optimal doses of warfarin (n = 123) were analyzed for warfarin and 7-hydroxywarfarin levels by HPLC, and CYP2C9*2 and CYP2C9*3 polymorphisms by PCR-RFLP.
- A multiple linear regression model was developed to study the impact of demographic and genetic variables on the pharmacokinetics of warfarin.
- *In silico* approaches were used to study the interactions between CYP2C9 and warfarin in the presence and absence of the variant allele.

Results & conclusion

- Multiple linear regression model showed that CYP2C9 polymorphisms are associated with higher warfarin/7-hydroxywarfarin ratio.
- *In silico* data revealed that warfarin forms three hydrogen bonds with CYP2C9*1 (two with protein backbone, one with protein side chain) and one π - π bond with F100.
- In CYP2C9*2 and CYP2C9*3 variants, two hydrogen bonds with protein backbone are disrupted. In double variant, all the hydrogen bonds are disrupted.
- In CYP2C9*2 and CYP2C9*3 variants, the distance between the C7 of S-warfarin and Fe-O was found to be increased resulting in reduced binding affinity.

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